Voyage economics and the time charter equivalent

What a dry bulk or tanker voyage earns a day, the freight it needs, the costs that decide it, and the physical fixture against the FFA, on the Baltic Exchange's routes in 2026

Research note by Nathan Couturier, 29 October 2026. Interactive site: https://nathancouturier.github.io/voyage-tce-model/. Code and data: https://github.com/nathancouturier/voyage-tce-model.

Summary

- A VLCC on TD22, Galveston to Ningbo, at the Baltic's assessed USD 58.22 million in the week to 23 October 2026, earns an estimated USD 463,639 a day over a 111.1 day round voyage; freight of USD 17.34 a tonne would only cover the voyage costs priced. The Baltic's own TCE that week was USD 451,905; the engine reads 2.6 percent higher, partly the port charges it leaves out.
- On routes outside Europe, bunkers are every sourced dollar of voyage cost at the August 2026 market; on voyages into or out of the EEA, EU allowances at the 2026 phase in are 12.6 to 17.4 percent of it (EUA EUR 82.31 a tonne). A Hormuz war risk premium of 7.5 percent of a USD 157 million VLCC hull would cost the charterer USD 11.78 million for each seven days inside the Gulf, 6.5 times TD3C's bunker bill.
- Round the Cape of Good Hope an LR2 on TC20 sails 31.5 more days than by Suez and needs USD 11.61 million (USD 128.94 a tonne) more freight, before Suez tolls, to earn what it earns by Suez, where the Red Sea premium on the laden transit costs its charterer USD 4.72 a tonne; Suez stays the better way for the owner until its tolls for the 2 transits together pass USD 7.00 million.
- For a VLCC from the Gulf a 7.5 percent premium costs the charterer USD 43.61 a tonne for one seven day period; 20 days of waiting inside the Gulf make 4 periods, USD 174.44 a tonne, and the owner needs USD 99.86 a tonne more freight to keep the TCE of the week to 23 October 2026.
- In the week to 3 July 2026 a C3 cargo at USD 30.11 a tonne earned USD 37,294 a day grossed up to the 5TC's gross hire (USD 35,429 net), between USD 419 and 4,060 a day above the 5TC FFA over its days, by which of two prints of the 29 June curve is read; C5 sat between USD 10,109 and 14,961 a day above it. By 23 October 2026 the C3 voyage sat USD 6,799 a day below the curve.
- Higher freight raises the speed that maximises the TCE and dearer bunkers lower it, on every one of the routes tested (C5, C3, TD22, TC20); for the tankers only at the lowest public freight, their best speed otherwise at the top of the range.

A TCE here is an estimate of what a voyage earns at stated inputs, not a forecast, and it is set against operating and capital costs, never net of them.

Of USD 58.22 million of freight, USD 51.53 million is left after commission and USD 4.51 million of voyage costs priced, all of them bunkers; port charges, with no public figure, are left out. Over 111.1 days (54.9 in ballast, 51.2 laden and 5.0 in port and waiting), the result is a TCE of USD 463,639 a day.

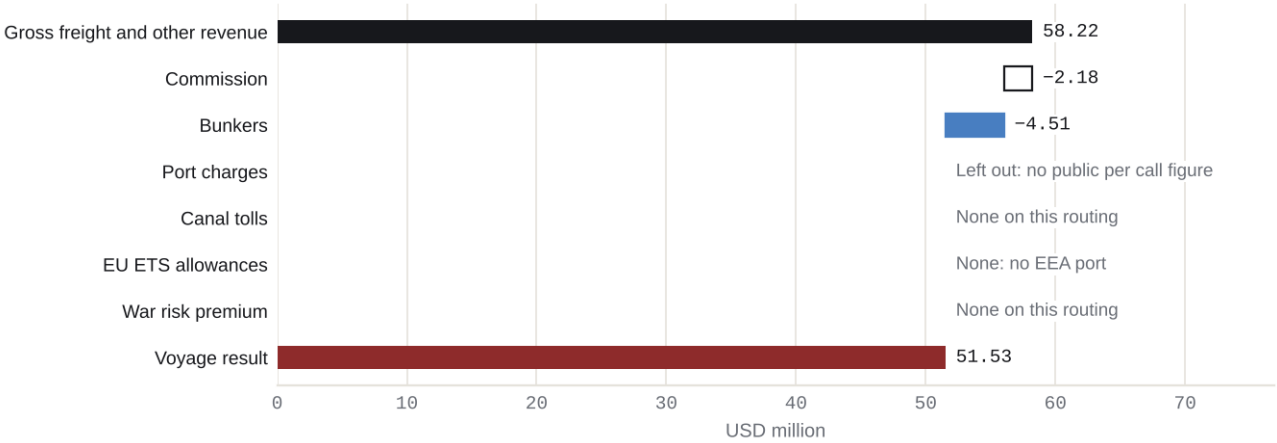


Figure 1. Engine output for TD22 in the week to 23 October 2026. Freight: Baltic Exchange via Hellenic Shipping News. Bunkers: Star Asia Shipbroking weekly market report, week 43 of 2026, 23 October 2026. A cost with no public figure is left out and named in its row. Sources retrieved 29 October 2026.

1. The question and the method

For a given ship, route and market: what does the voyage earn a day, what freight does it need, which costs decide the answer, and is fixing the cargo better than hedging it on paper? The routes are the Baltic Exchange's (Guide to Market Benchmarks v8.8, September 2026): C3, C5, P7 and the S4A trip for dry bulk; TD3C, TD22, TD20 and TD25 dirty; TC2, TC14, TC1 and TC5 clean; TC20 and TD34 where needed.

TCE = (freight + other revenue − commission − voyage costs) / voyage days, over the whole round voyage, the ballast leg included. Voyage costs are bunkers by sulphur zone (MGO inside emission control areas, measured on the route line), port charges, canal tolls, EU allowances and, where the owner pays it, the war risk premium. The tanker TCE follows the Baltic's own method (Appendix 3 of the guide): a 5 percent weather margin on steaming days, the round voyage from the discharge port, the port and waiting days and consumptions of the route's description, and bunkers at the hub nearest the load port; C3 alone is priced at Singapore, about 270 nm off its ballast leg, since the hub nearest Tubarao lies off it. The guide leaves a dry route's weather allowance to its panellists, so the same 5 percent on the dry routes is the study's choice. Worldscale freight is WS / 100 x flat rate, priced with the reader's own licensed flat rate. Commission is 5 percent on dry routes and 3.75 percent on tankers, address commission and brokerage together.

Every input is sourced, derived by a stated rule, the study's choice, left out as the Baltic leaves it out (a war risk premium) or empty: a figure with no public source, such as port charges and canal tolls, stays out of the TCE and the result names it. A dated market reads no price from after its date. The engine is a Python package, mirrored in the browser and held to it within one part in a billion on 295 cases.

2. Validation

The Baltic publishes its tanker routes on a Worldscale and a TCE basis. Only two study routes are assessed as a lump sum in dollars, TD22 and TC20, so only they can be priced without a licensed flat rate: 13 weeks between 12 December 2025 and 23 October 2026. With bunker prices first printed in the week, the engine reads 2.6 to 4.5 percent above the Baltic's TCE, and 1.8 to 4.7 percent with prices up to a week old, always above it. The gap grows with freight, from USD 4,213 a day at a USD 13.68 million lump sum (23 January 2026) to USD 11,653 at USD 79.61 million (9 October 2026), so the port charges the engine leaves out, a fixed sum a voyage, explain only part of it. The weeks to 13 March 2026 and 20 March 2026, just after the closure of Hormuz, read 17.3 and 9.3 percent above with tables of the same weeks, and why is not established; TD3C, priced at Fujairah, shows no such jump in those weeks (below), which points to TD22's Houston prices or to the hub the Baltic priced TD22 at before its guide rewrote the fuel price text in May 2026.

The engine against the Baltic's TD22 and TC20 TCEs, week by week: 1.8 to 4.7 percent above with bunker prices up to a week old, the two weeks after the closure of Hormuz aside.

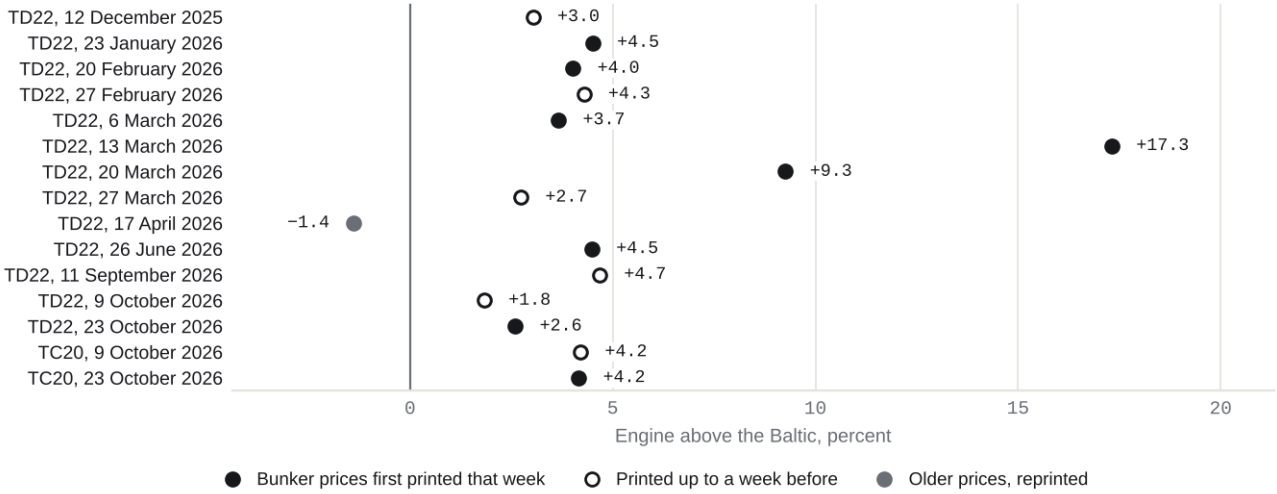


Figure 2. Baltic Exchange weekly tanker reports, weeks to 12 December 2025 to 23 October 2026, as republished by Hellenic Shipping News; bunkers: Star Asia Shipbroking weekly tables, a reprinted row dated by its first printing; engine output. Sources retrieved 29 October 2026.

The bunker table matters too: Star Asia sometimes reprints a week's table (the week to 17 April 2026, priced with Houston prices first printed on 20 March 2026, reads 1.4 percent below), and for 6 March 2026 Intermodal's Houston VLSFO, USD 689.30, against Star Asia's USD 584 moves the comparison from 3.7 to 1.0 percent. No public source states a flat rate for a study route's ports, but Platts printed its Persian Gulf to China rate in both units in January 2026

(w53.5 and USD 11.07 a tonne, w93 and USD 19.24), which implies USD 20.69 a tonne for its route, whose ports it does not state. At that rate the Baltic's TD3C weeks of 2026 read −1.9 to +10.9 percent against its TCE, the two weeks after the closure within that range. The other 52 Worldscale anchors need a licensed flat rate and are recorded, not tuned. The Baltic's worked TD3C calculation of July 2017, the one fully worked public example found, is reproduced line by line: laden 19.85 days, ballast 20.64, 46.49 in all, 1,389.20 t of fuel laden; it prints no prices, so no dollar figure is checked against it. Searoute distances fall within 2 percent of NGA Pub. 151 for 20 of 25 port pairs; the largest miss is Port Hedland to Busan (+4.2 percent).

Early March 2026: Platts' USD 62.07 a tonne for the Gulf to China on 2 March 2026 gives USD 319,741 a day with the bunkers of the week of 27 February 2026, and USD 255,921 with the early March premium of 2.5 percent of a USD 120 million hull in the owner's costs; the USD 89.00 that Oilprice.com reported on 3 March 2026 names no primary assessment and would give USD 468,621. The Baltic's TD3C TCE was USD 209,550 in the week to 27 February 2026 and USD 485,959 in the week to 6 March 2026.

3. Where the money goes

At one common dated market, August 2026 (the EU allowance at EUR 82.31 a tonne and 1.1594 dollars a euro), sourced voyage costs are bunkers and, on voyages that touch the EEA, EU allowances. The share of MGO in the bunker bill follows the miles inside emission control areas and the days at EU berths.

Table 1. Sourced voyage costs at the August 2026 market, by route.

Route	Voyage	Days	Sourced voyage costs, USD	Bunkers, %	EU ETS, %	MGO in bunkers, %
C3	Tubarao to Qingdao	76.5	2,746,360	100.0	0.0	0.0
C5	West Australia to Qingdao	29.3	881,274	100.0	0.0	0.0
P7	US Gulf to Qingdao grain	87.0	1,507,418	100.0	0.0	8.2
S4A	US Gulf trip to Skaw-Passero	42.6	806,116	84.6	15.4	35.6
TD3C	Middle East Gulf to China	47.0	1,816,997	100.0	0.0	0.0
TD22	US Gulf to China	111.1	3,969,107	100.0	0.0	10.4
TD20	West Africa to UK-Continent	36.3	1,150,605	82.6	17.4	31.0
TD25	US Gulf to UK-Continent	40.0	1,268,237	84.5	15.5	48.1
TC2	Continent to US Atlantic coast	28.3	605,358	87.4	12.6	69.9
TC14	US Gulf to Continent	40.7	814,501	84.7	15.3	46.4
TC1	Middle East Gulf to Japan (LR2)	51.4	1,221,620	100.0	0.0	0.0
TC5	Middle East Gulf to Japan (LR1)	51.9	1,100,833	100.0	0.0	0.0

Engine output at the common market: bunkers, Ship & Bunker's August and third quarter averages at each route's hub; allowances, DEHSt's August average at the 2026 phase in; dollars, the Federal Reserve's H.10. Port charges and canal tolls have no public figure. S4A's row holds the charterer's trip costs and the owner's ballast leg. Sources retrieved 29 October 2026.

4. Speed

The TCE was computed on a 0.01 knot grid from 1 knot below the slowest published speed to 1 knot above the fastest, at the lowest, median and highest public freight of the weeks studied (December 2025 to October 2026) and at the lowest weekly, the August 2026 and the highest weekly bunker price at the route's hub. Along every line of that grid the best speed never falls as freight rises or rises as bunkers rise, and somewhere it moves: the textbook result holds on every route tested. Tanker consumption comes at two speeds, so the fitted exponent is low and the best tanker speed sits at the top of the range except at the lowest freight.

Table 2. Laden speed that maximises the TCE, knots.

Route	Low freight	Median freight	High freight	Cheap bunkers	Dear bunkers	Fitted exponent
C5	12.0	12.0	13.0	15.0 (top)	10.5	2.43
C3	10.7	12.0	13.0	15.0 (top)	10.0 (bottom)	2.43
TD22	13.9	14.0 (top)	14.0 (top)	14.0 (top)	14.0 (top)	2.18
TC20	14.0 (top)	14.0 (top)	14.0 (top)	14.0 (top)	14.0 (top)	1.60

Freight columns at the common market's bunkers, bunker columns at the median freight; top, bottom: an end of the range searched. Engine output on the guide's speed pairs; freight: the Baltic's reports and the dry weeks' rates; bunkers: Star Asia, Ship & Bunker. Sources retrieved 29 October 2026.

5. Detours: Suez, the Cape and Hormuz

Asia to Europe. An LR2 on TC20, Jubail to Rotterdam at the Baltic's lump sum of USD 21.51 million in the week to 23 October 2026, earns the owner USD 342,369 a day by Suez before any toll and USD 210,173 round the Cape, 31.5 days and 0.43 billion tonne miles more. The charterer pays the Gulf's premium on both routings and the Red Sea's on the laden Suez transit, USD 4.72 a tonne; the ballast transit, the approach to the next loading, is the owner's (Gard, March 2026) and comes off the Suez TCE. A Panamax one way from Qingdao to Rotterdam, like the P4 trip, needs USD 29.01 a tonne by Suez to earn P4's hire net of commission (USD 12,878 a day), USD 34.54 by Panama and USD 36.08 round the Cape, before port time, port charges and tolls; at the Cape's freight, Suez could bear USD 443,218 of tolls and waiting, Panama USD 96,657.

Hormuz. A TD3C voyage spends 6.5 days inside the Gulf, one seven day period of premium; each further week of waiting adds a period for the charterer and the TCE it forgoes for the owner. Loading off Fujairah, the export VLCC sails 2.7 fewer days, but the shuttle through the strait was reported at USD 16 to 20 a barrel (about USD 118.52 to USD 148.15 a tonne) and at USD 50 million a transit, and the transfer's own cost has no public figure. The Baltic assessed TD34, a cargo loading outside the strait, at USD 656,587 a day against TD3C's USD 1,287,851 in the week to 23 October 2026, which the engine turns into freight of USD 114.38 a tonne against USD 241.39.

Per tonne of cargo, the Cape costs an LR2's owner far more freight than the Red Sea premium costs its charterer; for a VLCC loading in the Gulf, each further week inside adds a period of premium and a week of lost earnings.

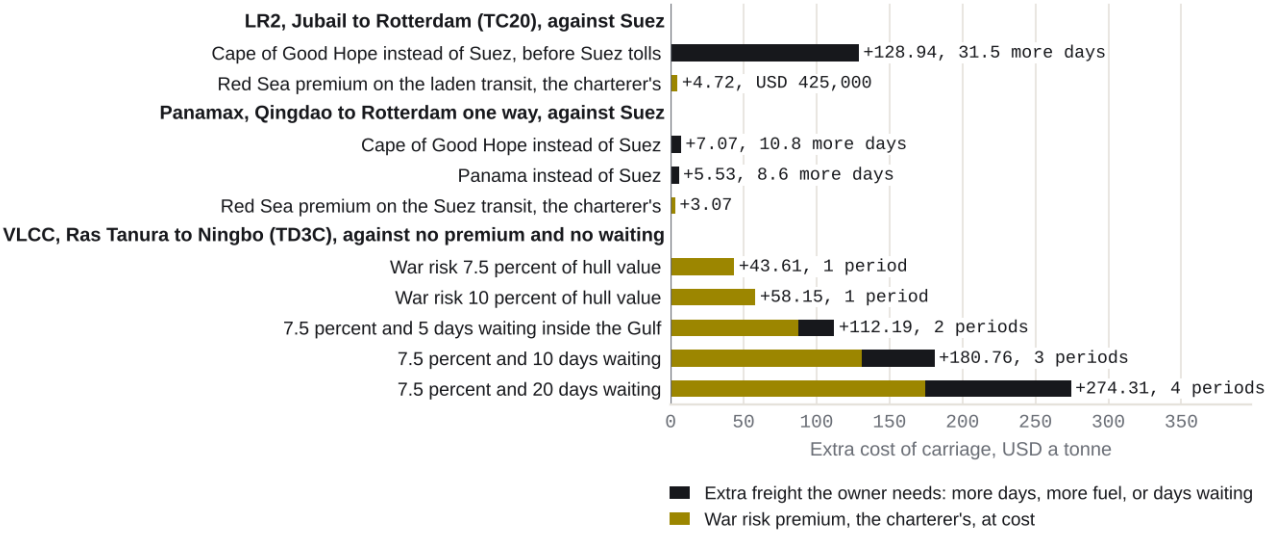


Figure 3. Engine output, week to 23 October 2026: TC20 at the Baltic's lump sum, the Panamax one way at the freight that earns the P4 hire net of its commission, TD3C at the Baltic's TCE of the week (USD 1,287,851 a day). War risk: Marsh, 22 July 2026, on Xclusiv Shipbrokers' hull values of 7 September 2026. Suez and Panama tolls and the crew war bonus have no public figure for these ships and are left out; the transfer chain and TD34 are set out in section 5. Sources retrieved 29 October 2026.

6. Carbon

EU allowances cover half the emissions of a voyage into or out of the EEA and all of one between two EEA ports and within a port of call, for 40 percent of 2024 emissions, 70 percent of 2025 and 100 percent from 2026, with methane and nitrous oxide (Directive 2003/87/EC, Articles 3ga and 3gb, Annex I). On a time charter the charterer provides the allowances for the charter's emissions (BIMCO ETS Allowances Clause, 2022) and the owner surrenders them (Article 3gc): on the S4A trip the charterer pays USD 66,690 and the owner USD 57,491 for the ballast leg back. FuelEU's penalty is assessed on a ship's year, so it stays out of the TCE.

Table 3. EU ETS cost of one voyage by phase in year, at the August 2026 allowance price.

Voyage	2024	2025	2026	2026, USD a tonne of cargo
TC23, between two EEA ports	USD 11,310	USD 19,793	USD 28,708	0.96
TD20, into the EEA	USD 78,791	USD 137,885	USD 200,059	1.54
TD3C, outside the EEA	USD 0	USD 0	USD 0	0.00

Engine output at EUR 82.31 a tonne (DEHSt (German auctions, proxy), monthly average for August 2026) and 1.1594 dollars a euro (Federal Reserve H.10, mean of the 21 published days of August 2026); emission factors of Regulation (EU) 2015/757 Annex I as amended, global warming potentials 28 for methane and 265 for nitrous oxide. Sources retrieved 29 October 2026.

7. Physical against paper

5TC FFAs settle on the month's average of the daily index in dollars a day (EEX's C5TM on the 182,000 t ship), C3, C5 and the ICE tanker routes on the month's average in dollars a tonne (TC5 and the 75,000 t Arab Gulf to Japan contract on Platts, the rest on the Baltic). A voyage's net TCE is grossed up by its commission before it meets gross hire (guide, 7.4.3), here the FFA strip over the voyage's days from the fixture date; the 29 June curve was printed twice (July USD 31,367, then 36,232). The C3 voyage of the week to 3 July 2026 is hedged with 76.5 days of 5TC (72.7 hedge its net earnings) or 170,000 tonnes of C3; that day's spot 5TC, USD 33,678, is the 180,000 t basis, so the 182,000 t ship meets USD 37,181. A C5 cargo fixed on 22 October 2026 at USD 14.15 a tonne for 5 to 7 November 2026 matched the route that day in the report's narrative, USD 14.15; its table and narrative differ by USD 0.10 at the next close, about USD 552 a day.

Grossed up by their commission, the C5 voyages earn well above the 5TC FFA over their days in both weeks; C3 sat between USD 419 and 4,060 above it in July, depending on which print of the 29 June curve is read, and USD 6,799 a day below it in October.

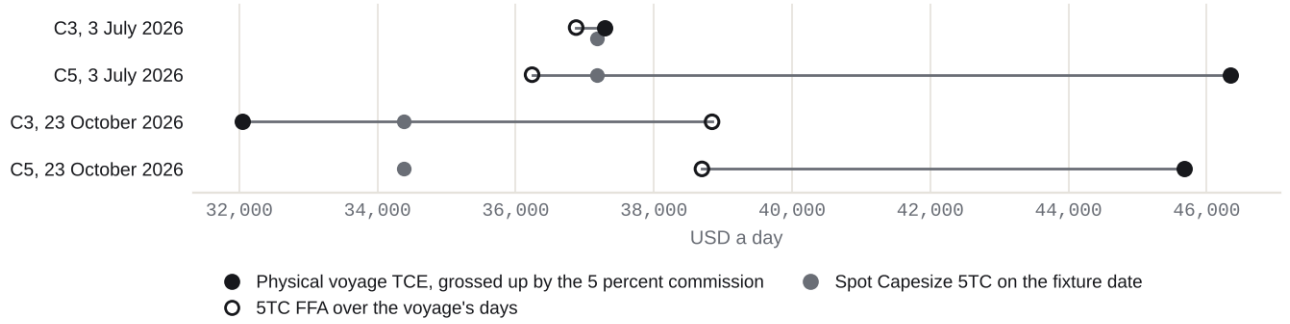


Figure 4. Engine output for C3 and C5 at the route rates and bunker tables of the fixture dates, bunkers at Singapore; FFA: Banchero Costa, Dry bulk FFA assessments, curves of 29 June 2026 and 25 October 2026; spot 5TC: Signal Group and Banchero Costa, the July figure on the 180,000 t basis and here on the 182,000 t. Sources retrieved 29 October 2026.

8. What the study cannot do

- Port charges have no public per call figure: every TCE leaves them out (each USD 100,000 is USD 1,307 a day on C3, USD 3,419 on C5 and USD 900 on TD22), and they explain only part of the gap to the Baltic.
- No public source states a Worldscale flat rate for a study route's ports, nor a fully worked voyage estimate with prices: Worldscale rates need the reader's own flat rate (TD3C's 2026 check aside), and the Baltic's 2017 TD3C note checks days and fuel only.
- Suez and Panama tolls, and canal time on the dry routings, are left out. Bulk carrier speeds and the 5 percent dry weather margin are the study's choices; without the margin the Capesize TCEs here read 6 to 11 percent higher.
- Bunker prices are broker tables and averages at four hubs, not what a ship paid, and two brokers can differ widely for the same day; the dry FFA curves are dated assessments, one date printed twice with different values, and no public tanker or route FFA quote in dollars a tonne was found.
- War risk premiums are the latest dated rates, from July 2026; the listed area is taken to end at the strait, a lower bound, though the Joint War Committee widened it in 2026.

Sources and reproduction

Baltic Exchange (Guide to Market Benchmarks v8.8; TD3C TCE calculation, July 2017), its weekly tanker reports, December 2025 to October 2026, and the brokers' and insurers' reports cited, via Hellenic Shipping News; Ship & Bunker; European Commission and DEHSt; Federal Reserve H.10; EUR-Lex and IMO; NGA Pub. 150 and 151; searoute 1.6.0 over Eurostat SeaRoute; EEX and ICE. Retrievals and hashes: data/raw/manifest.json; scripts/note.py writes this note.